

K.Sethum Sithmaka Perera

Cyber Security Undergraduate

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PROFILE

Self-Motivated, Highly Talented, and Energetic Individual, as well as a National-level Badminton Athlete, bringing discipline, resilience, and a strong competitive spirit. I am a Cybersecurity specialized student with solid knowledge of information security and network security principles, complemented by hands-on experience in Java Spring Boot projects, including backend development and RESTful APIs. I also have an academic background in Industrial Statistics and Mathematical Finance, strengthening my analytical and problem-solving abilities.

I am seeking an opportunity in a reputed organization where I can contribute with maximum commitment while continuously growing and adding value to organizational objectives.

Professional Experience

- 08/2024 – 05/2025

Athlete, Sri lankan Army
 - Selected and contracted to represent the Sri Lanka Army in competitive badminton.
 - Competed at national-level tournaments and training camps.
 - Maintained peak athletic performance through structured training programs.
 - Upheld professional conduct within a military organization.
- 2019 – 2024

Athlete, Sri Lanka Badminton Association
 - Represented Sri Lanka in national and international badminton competitions
 - Maintained elite-level training and performance standards for 5 years
 - Demonstrated discipline, time management, and competitive excellence
 - Collaborated with coaches and team members to achieve performance goals
 - Balanced academics with intensive training schedules

Education

- 2024

Bsc(Hons) Information Technology specialising in Cyber Security,
Sri Lanka Institute of Information Technology
 - Grade : 3.5/4.0
- 2025

BSc (Hons) in Industrial Statistics & Mathematical Finance,
University of Colombo
 - Reading
- 2023

GCE Advance Level, Maths Stream, D.S Senanayake College Colombo 07
 - Physics - B
 - Chemistry - B
 - Combined Maths - C
- 2020

GCE Ordinary Level, D.S Senanayake College Colombo 07
 - 9A's

Skills

Programming Languages

- Java
- JavaScript
- Python
- C
- R

Project Collaboration & Workflow

- Git
- GitHub
- Agile Methodologies

Backend Frameworks

- Spring Boot

Web Technologies

- HTML
- CSS

Software Development Practices

- OOP
- Debugging
- RESTful API Development

Databases

MySQL

Machine Learning & AI

- Machine Learning
- Supervised Learning
- Model Training & Evaluation
- TensorFlow
- PyTorch
- Jupyter Notebook
- Scikit-learn
- Matplotlib
- Pandas

Tools & Development Environments

- PyCharm
- Arduino IDE
- LibreOffice
- GNU Octave

Certificates

- Python for Beginners - University of Moratuwa (2026 Jan)
- Python Programming - University of Moratuwa (2026 Feb)
- AI/ML Engineer - Stage 1 SLIIT (2026 Jan)
- AI/ML Engineer - Stage 2 SLIIT (2026 Jan)

Projects

2025

Driving School Registration and Scheduling System

Technologies

**Java, Spring Boot, MySQL, JavaScript, HTML, CSS*

- Developed a web-based Driving School Management System to handle student registration and lesson scheduling.
- Implemented booking, cancellation, and schedule management functionalities.
- Built full CRUD operations (add, update, delete, view) for students, instructors, and lesson records.
- Designed feedback management module for students.
- Applied Object-Oriented Programming (OOP) principles including encapsulation and inheritance.
- Designed class relationships and modular system architecture using Spring Boot framework.
- Implemented backend REST APIs and integrated with frontend using HTML, CSS, and JavaScript.

2024

Smart Water Tank Monitoring System

Technologies

**C++, Arduino, Ultrasonic Sensor, Embedded Systems Design*

- Designed and implemented a Smart Water Tank Monitoring System using Arduino for real-time water level monitoring.
- Integrated an ultrasonic sensor to measure water levels accurately.
- Developed embedded C++ logic to automate water pump control based on water level thresholds.
- Implemented automatic motor ON/OFF functionality to prevent overflow and dry running.
- Added visual alert indicators for low water levels.
- Designed the overall system architecture and circuit diagram for efficient hardware-software integration.
- Demonstrated the feasibility of IoT-based smart water management solutions.

2025

Web-Based Bus Scheduling and Booking System

Technologies

**Java, Spring Boot, MySQL, JSP, HTML, CSS, Data Structures*

- Developed a web-based Bus Scheduling and Booking System for managing bus schedules and reservations.
- Implemented booking, cancellation, and schedule viewing functionalities.
- Built CRUD operations (add, update, delete, search) for bus, route, and booking management.
- Implemented an online payment feature for secure bus ticket booking.
- Designed and implemented backend logic using Java and Spring Boot.
- Applied data structures for efficient booking record handling.
- Integrated JSP with backend to dynamically display booking and schedule data.
- Developed feedback system for passengers.

2025

Student Depression Prediction System

Technologies

**Machine Learning Classification | Python · scikit-learn · pandas*

- Developed an end-to-end machine learning pipeline to predict depression risk among students using academic, financial, and lifestyle features

- Performed data preprocessing including missing value imputation, feature encoding, and exploratory data analysis on a real-world student mental health dataset
- Trained and evaluated 6 classification models: Logistic Regression, K-Nearest Neighbors, Random Forest, Gradient Boosting, Gaussian Naïve Bayes, and Decision Tree
- Achieved best F1 Score of 0.871 and AUC of 0.916 using Gradient Boosting; conducted full model comparison across Accuracy, Precision, Recall, and F1 metrics
- Visualized insights and model performance using Matplotlib and Seaborn to support data-driven conclusions

2026

Multi-Client TCP Authentication Server

Technologies

*C · Python · POSIX Sockets · Linux

Built a concurrent TCP authentication server in C handling multiple simultaneous clients using fork() with SIGCHLD-based zombie process prevention.

- - Implemented salted password hashing (djb2 algorithm) — no plaintext credentials stored
- - Token-based session management with automatic 300-second expiry per client
- - Account lockout after 3 failed login attempts with a 5-minute cooldown
- - Per-client rate limiting of 5 commands per 60 seconds with ERR 429 enforcement
- - Custom LEN:<n> framed TCP protocol for reliable message boundary handling
- - Timestamped audit logging capturing IP, PID, username, command and result
- - Python CLI client built to interact with and test the server

2025

Web-Based Trip Management System

Technologies

*Java, Spring Boot, MySQL, JSP, JSTL, HTML, CSS, Spring Data JPA

- Developed a web-based Trip Management System to handle trip bookings, vehicle assignments, guide scheduling, and passenger feedback.
- Implemented booking and cancellation functionalities with full CRUD operations for booking, vehicle, and user management.
- Designed and integrated an online payment feature to support secure trip reservations.
- Built vehicle assignment and guide scheduling modules to efficiently manage trip resources.
- Developed backend logic using Java, Spring Boot, and Spring Data JPA following MVC architecture.
- Integrated JSP and JSTL with the backend to dynamically render trip and booking data on the frontend.
- Implemented a user management module with registration and profile handling.
- Developed a feedback management system to collect and manage passenger reviews.

SPORT ACHIEVEMENTS

BADMINTON

International Level

- Silver medalist in under 21 South Asian game - Maldives [2019]
- Player of Under 19 World Junior Championship (Captain) - Spain [2022]
- South Asian Region Badminton Coaching Camp - India [2018]
- Player of Under 15/17 Asian Junior Championship (Under 15 captain) - Myanmar [2018]

National Level

- 49 National Sports (2025) Team - Champions
- North Central Province Open Badminton Championship (2024) - Open Men's Singles Runners up
- Western Province open Badminton Champions (2024) - Open Men's Singles Runners up
- 48 National Sports (2024) Team - Champion
- U 15, U 17 and U19 Boys Singles Ranking No 01
- Western Province Open Badminton Championship (2020) - U 17 Boys Singles (Champion)
- SSC Open Badminton Championship (2019) - U 17 Boys Singles (Champion)
- Awarded Western Province School Colors for Badminton (2019)
- Sri Lanka LINING Shuttle League – Team Event (Champions) (2020)
- 69th Nationals Badminton Championship 2021 - U 19 Boys Singles (Champion)

University and School Level

- Participated in School Teams
- Under 20 All Island School Games - Champions [2023] – School - Team captain
- Under 20 All Island School Games – 2nd Runner up[2022]
- Under 17 All Island School Games - Runner up [2019]
- School Colors 2014 to 2023
- Participated in University Teams
- Silver Medalist - Sri Lanka University Games - SLUG (Team Event) – University of Colombo
- Champions – APIIT Extravaganza 2025 – SLIIT University
- SLIIT University Colours - 2024
- Champions – ICBT Smash Down 2024 – SLIIT University

Awards

2021	69th Nationals Badminton Championship , Sri Lanka Badminton Upcoming Player
2020	Sri Lanka LINING Shuttle League , Sri Lanka Badminton Most Valuable Junior Player
2016	School Colors , D.S Senanayake College Upcoming Player

References

Mrs.Hasalika Siriwardana, *Teacher in charge badminton*, D S Senanayaka Collage
0773868455